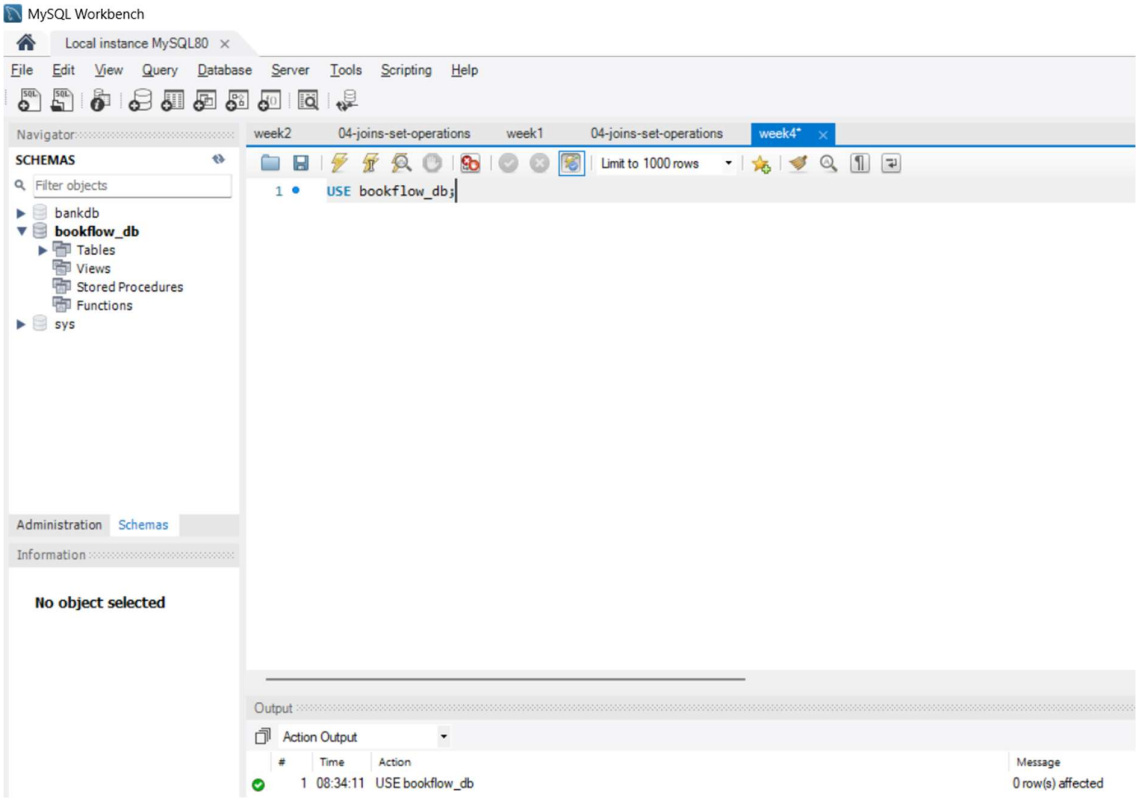
WEEK – 4

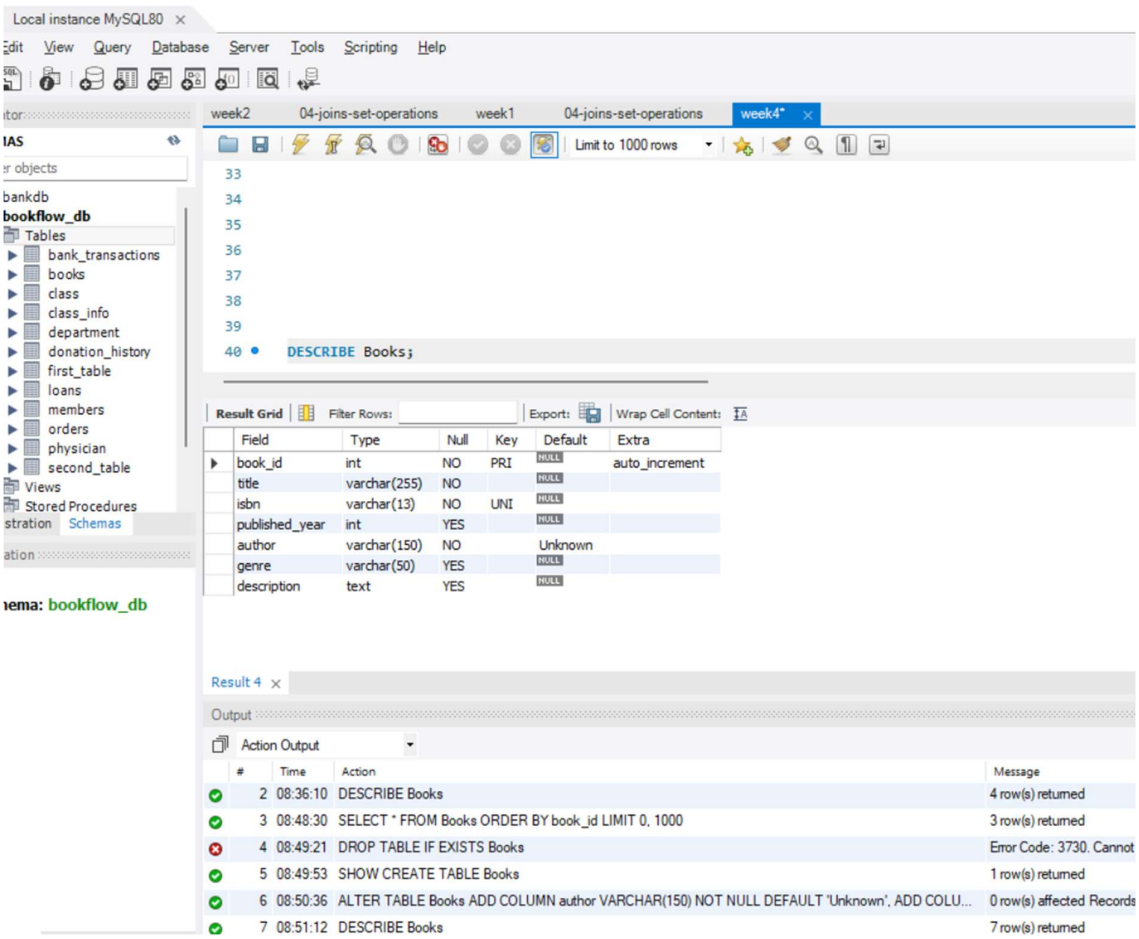
Mohammad Rehan

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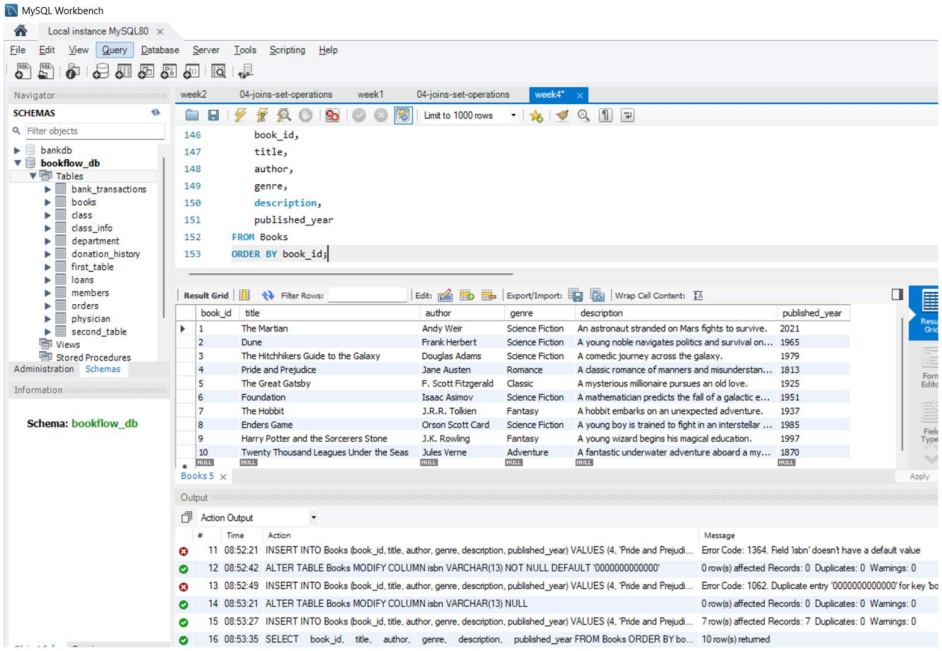
1) Create Database



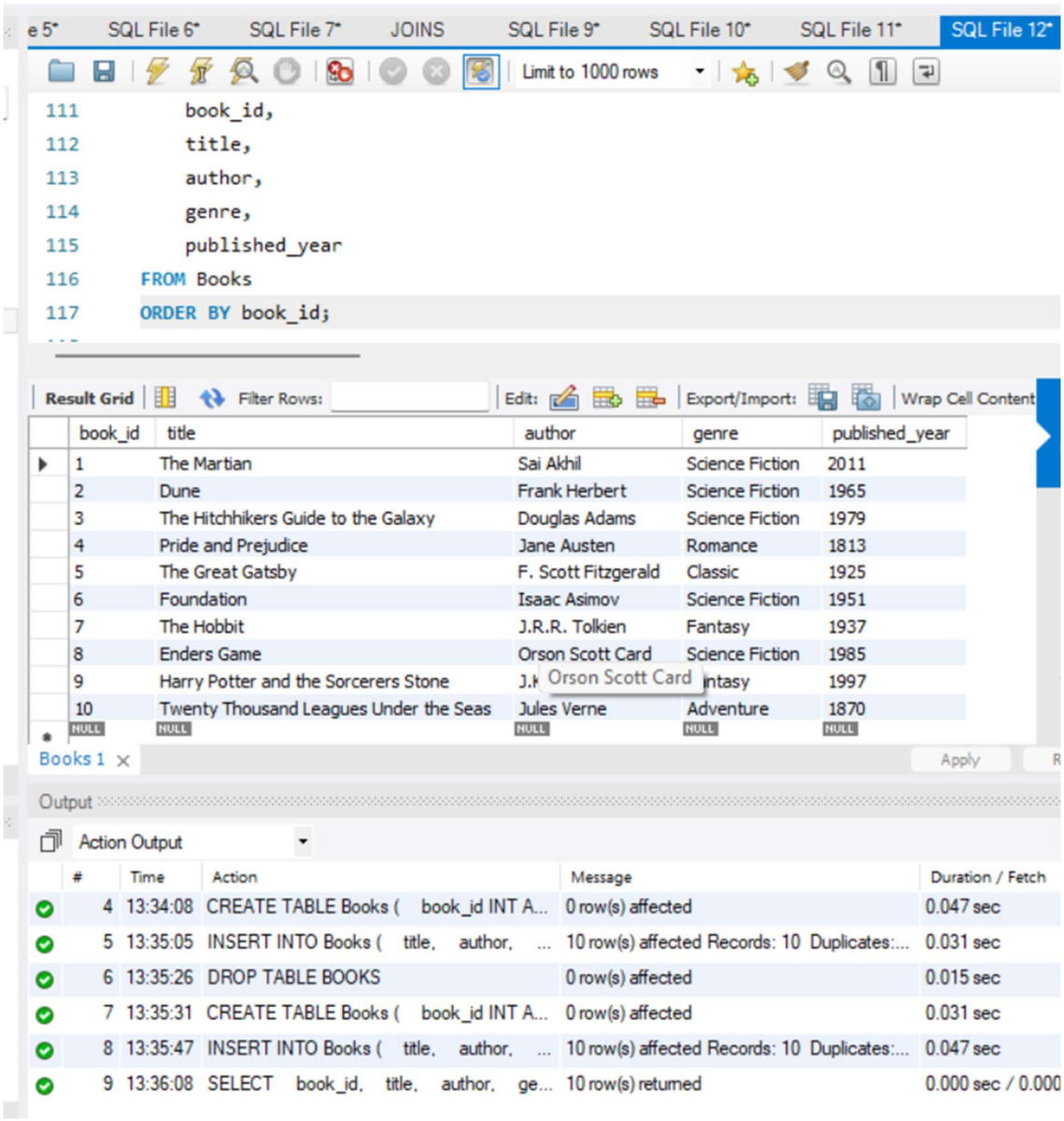
2) CREATE BOOKS TABLE



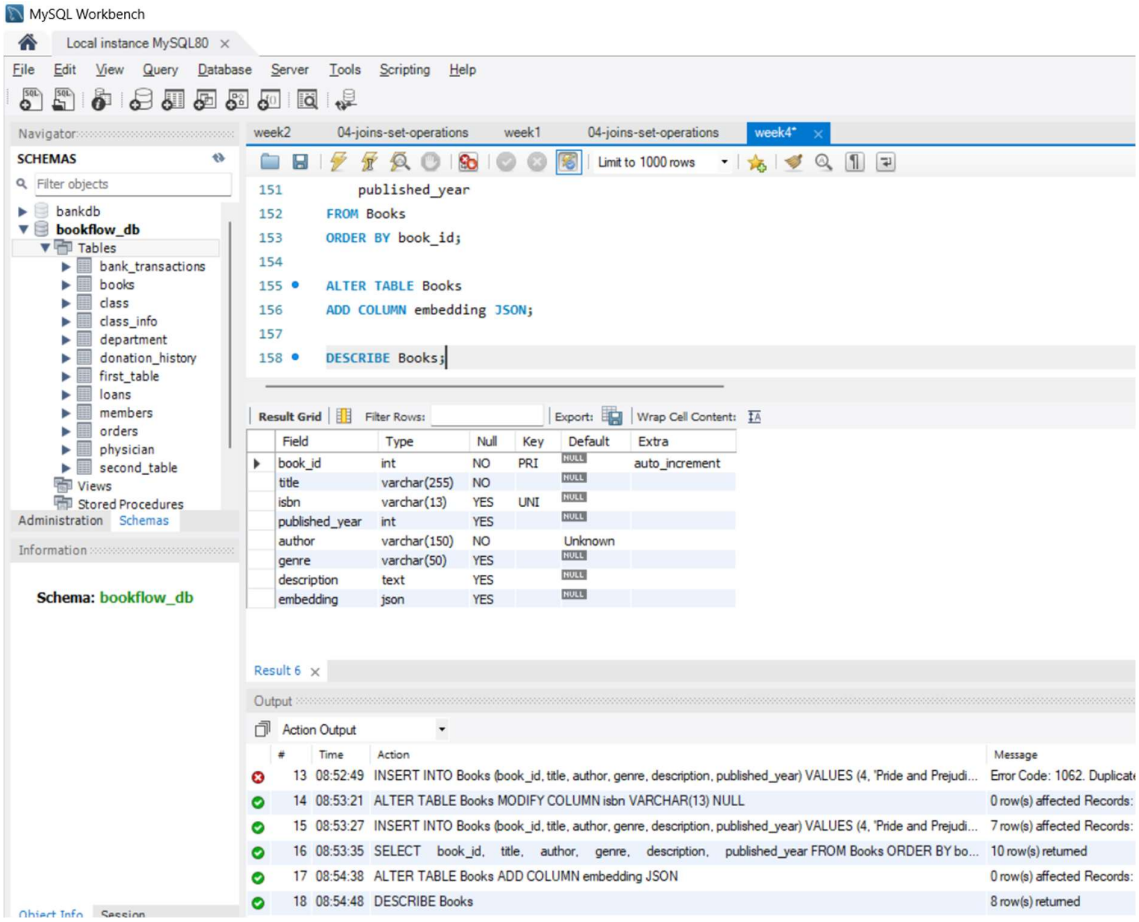
3) INSERT BOOK DATA



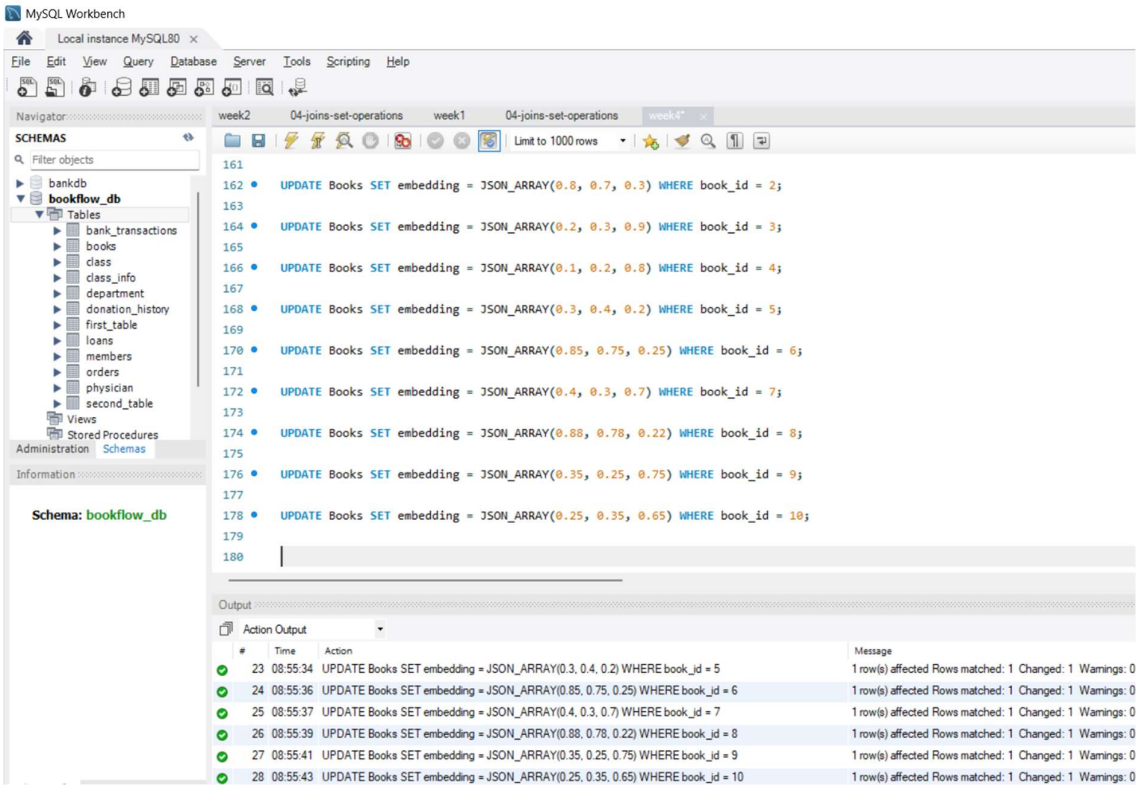
4) Display Books



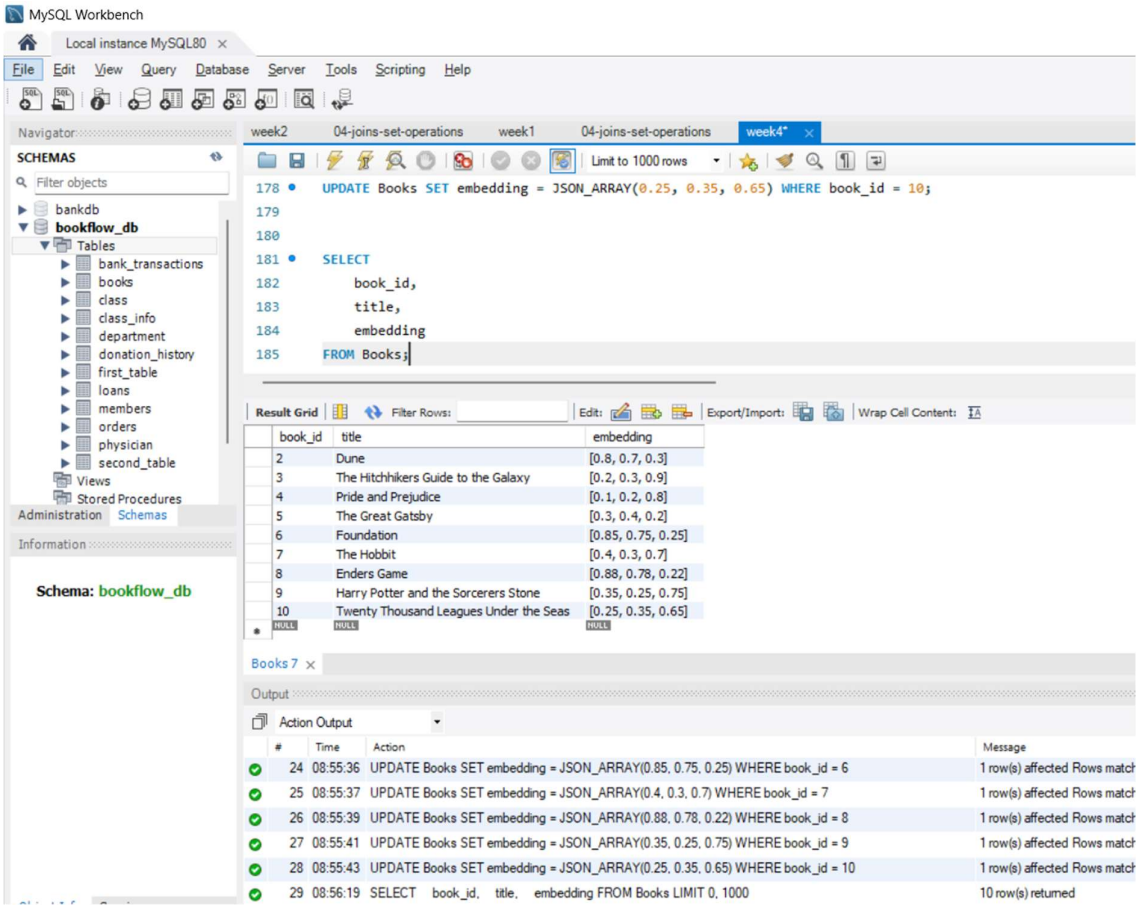
5) Add Embedding Column



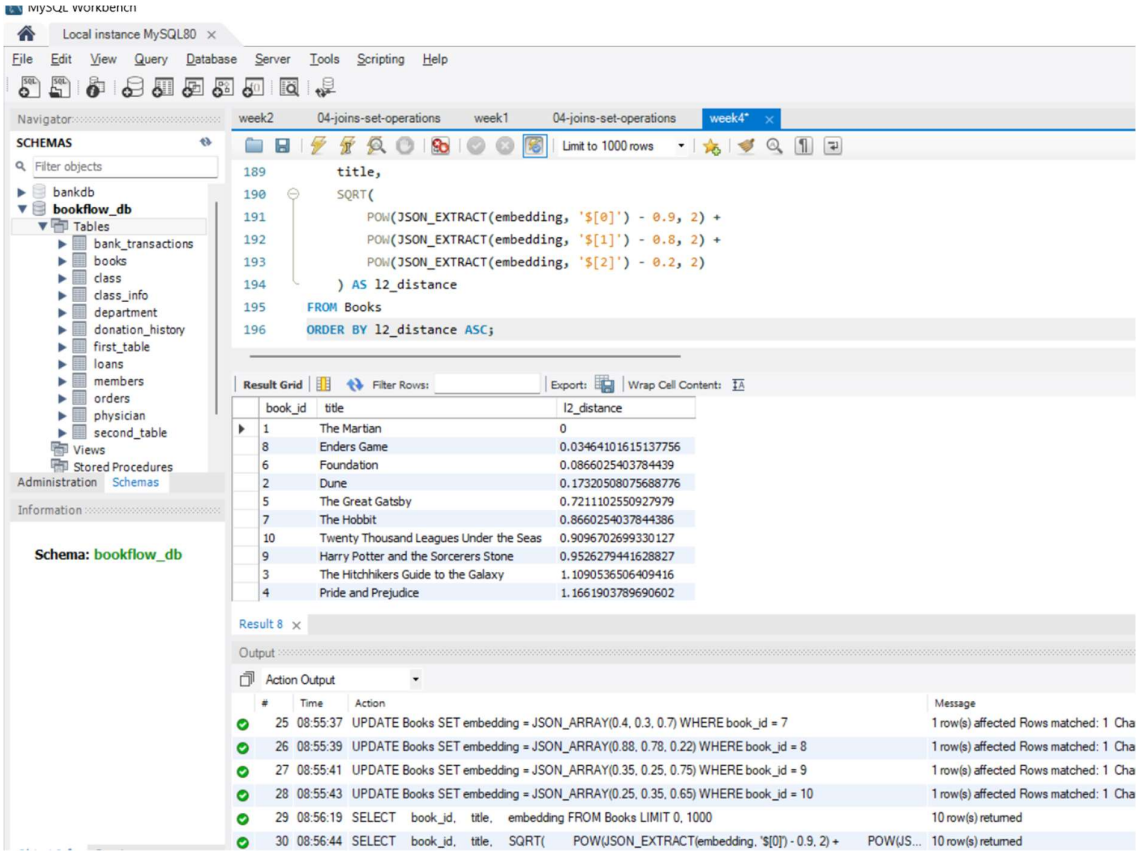
6) Insert Mock Vectors



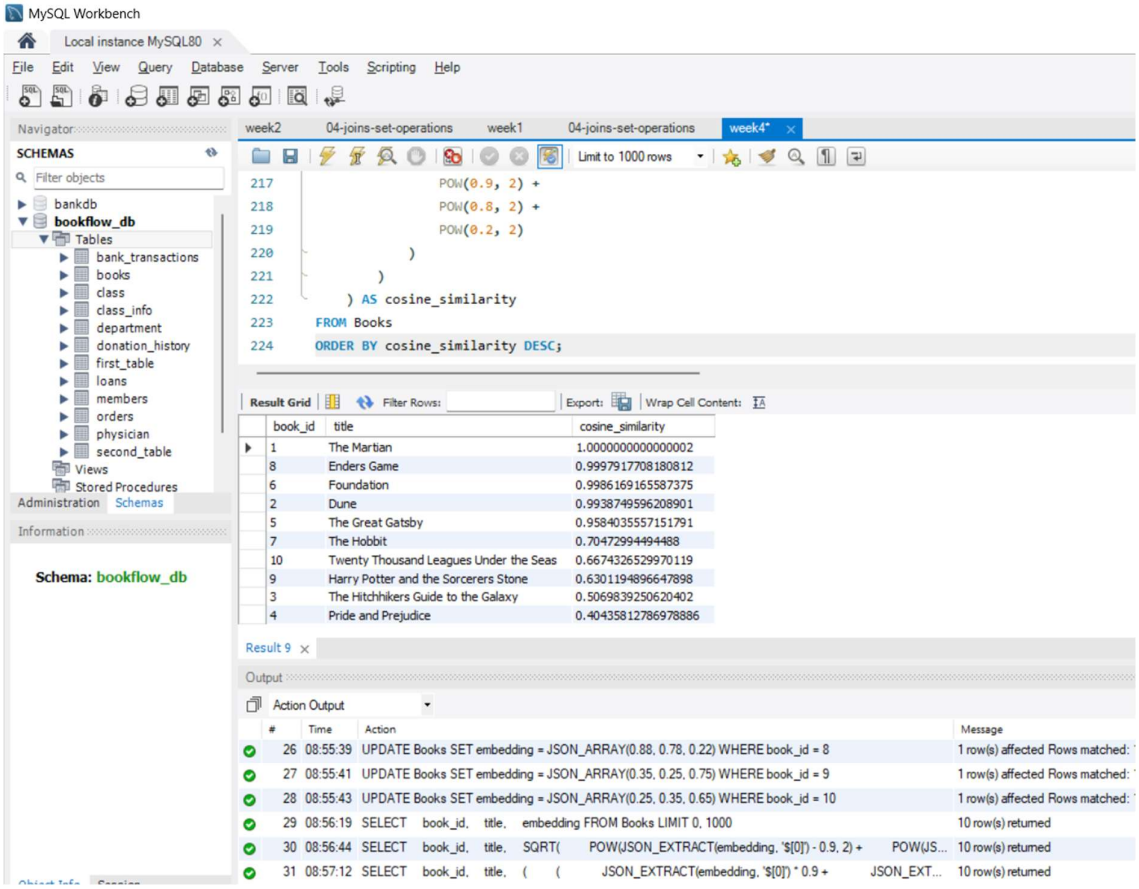
7) Verify all Vectors



8) L2 Distance



9) Cosine Similarity


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
bankdb
bookflow_db
Tables
bank_transactions
books
class
class_info
department
donation_history
first_table
loans
members
orders
physician
second_table
Views
Stored Procedures
Administration Schemas
Information
Schema: bookflow_db
week2 04-joins-set-operations week1 04-joins-set-operations week4*
Limit to 1000 rows
POW(0.9, 2) +
POW(0.8, 2) +
POW(0.2, 2)
) AS cosine_similarity
FROM Books
ORDER BY cosine_similarity DESC;
Result Grid Filter Rows: Export: Wrap Cell Content:
book_id title cosine_similarity
1 The Martian 1.0000000000000002
8 Enders Game 0.9997917708180812
6 Foundation 0.9986169165587375
2 Dune 0.9938749596208901
5 The Great Gatsby 0.9584035557151791
7 The Hobbit 0.70472994494488
10 Twenty Thousand Leagues Under the Seas 0.6674326529970119
9 Harry Potter and the Sorcerers Stone 0.6301194896647898
3 The Hitchhikers Guide to the Galaxy 0.5069839250620402
4 Pride and Prejudice 0.40435812786978886
Result 9
Output
Action Output
Time Action Message
26 08:55:39 UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.78, 0.22) WHERE book_id = 8 1 row(s) affected Rows matched:
27 08:55:41 UPDATE Books SET embedding = JSON_ARRAY(0.35, 0.25, 0.75) WHERE book_id = 9 1 row(s) affected Rows matched:
28 08:55:43 UPDATE Books SET embedding = JSON_ARRAY(0.25, 0.35, 0.65) WHERE book_id = 10 1 row(s) affected Rows matched:
29 08:56:19 SELECT book_id, title, embedding FROM Books LIMIT 0, 1000 10 row(s) returned
30 08:56:44 SELECT book_id, title, SQRT(POW(JSON_EXTRACT(embedding, '$[0]') - 0.9, 2) + POW(JS... 10 row(s) returned
31 08:57:12 SELECT book_id, title, ((JSON_EXTRACT(embedding, '$[0]') * 0.9 + JSON_EXT... 10 row(s) returned